theatre—in sanitised gowns, gloves, headwear and shoes. On the assembly line, robotic arms guide the gleaming iridescent silicon wafers reminiscent of compact discs of yore.

These discs are the building blocks of chip manufacturing. Each wafer holds thousands of chips etched in meticulous rows, fabricated in foundries abroad, using over 500 chemical, gaseous and electrical processes. A single chip is a beehive of transistors stacked in microscopic layers, with over 40 metres of micro-thin wiring threading the circuits. A whirring machine using diamond cutters slices the wafer into individual chips. They are then subjected to rigorous temperature, vibration and pressure checks, and scanned for material flaws before being packaged and dispatched to their destinations. To start with, CG Semi will process half a million chips daily and will scale it up to 15 million once its second complex is ready in 2027.



"Every chip we make is a step towards India's technological sovereignty. It shows how government and industry can come together with conviction, capital and scale, and deliver rapidly"

S. VELLAYAN | *Chairman, CG Power and Industrial Solutions*



LEARNING FROM FAILURES

What makes Modi's semiconductor mission even more challenging is India's six-decade track record of failure. In the 1960s, Robert Noyce, co-inventor of the integrated circuit and co-founder first of Fairchild Semiconductor and then of Intel, wanted to set up a unit in India but, confronted with red tape, settled for Hong Kong. Then, in 1983, Indira Gandhi announced the setting up of the Semi-Conductor Laboratory (SCL) in Chandigarh, complete with a chip foundry. But a devastating fire gutted the facility in 1989, and it could never recover or regain its expertise since.

In the 2000s, when Manmohan Singh was prime minister, he made several attempts to bring chip expertise to India, but failed. Bureaucratic roadblocks played spoilsport again when Intel in 2005 announced plans to invest $1 billion in a chip facility in Tamil Nadu, but shifted shop to China following restrictions on importing equipment. Two years later, Manmohan Singh made another attempt, announcing a Special Incentive Package Scheme (SIPS) to encourage electronics manufacturing, including semiconductors, in India, with an attractive 25 per cent capital subsidy. The 2008 global financial meltdown put paid to that ambition. The scheme was revived in 2012, with the government setting aside Rs 39,000 crore as a subsidy to construct two chip fabrication plants. Both IBM and the Chinese semicon giant HSMC participated in the bidding process, but eventually withdrew, citing an unfavourable market environment.

Learning from this string of failures, officials in the Modi government worked on how to sidestep these pitfalls. As Amitesh Sinha, the affable CEO of ISM, points out, earlier efforts failed because India went straight for the high end of the business—a semiconductor fabrication (fab) unit—a risky proposition in the absence of an ecosystem. This time, therefore, the Modi government is inviting projects across the entire spectrum of chip-making, from design, fabrication, assembly to testing and packaging. The financial incentive is also better. Instead of loans or equity stakes, the government is offering direct grants. The Centre is also adopting a *pari-passu* approach—for every rupee an investor spends, the government gives a matching grant. States have said they will bear another 20-30 per



"Failure is not an option for any of us. We must show the world that India can do it. But it's important that the government continue to support us for the next four years"

PANKAJ GULATI | *President, CDIL*